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1 1 Introduction

1.1 1.1 Purpose

Campus Stalls is a web-based student marketplace designed specifically for buying, selling, exchanging, and discovering products required for college life.

The platform provides a centralized environment where students can list and find useful items such as textbooks, lab coats, calculators, stationery, academic materials, accessories, and other permitted college-related goods.

This Software Requirements Specification (SRS) defines the functional and non-functional requirements of Campus Stalls. It serves as a common reference for developers, testers, project guides, administrators, and end users.

1.2 1.2 Document Conventions

The following conventions are used throughout this document:

- Main text uses 11 pt font.
- Section headings are displayed in bold.
- Functional requirements are identified as REQ-1, REQ-2, etc.
- Priority levels are Critical, High, Medium, and Low.

1.3 1.3 Intended Audience

This document is intended for:

- Project Guide
- Developers
- Test Engineers
- System Administrator
- Student Buyers and Sellers
- College/Marketplace Administrator

1.4 1.4 Project Scope

Campus Stalls provides a campus-focused marketplace where students can conveniently discover and exchange college-related products.

Users can:

- Register and securely log in.
- Create and manage student profiles.
- List new or used college-related products.
- Search and filter products by different criteria.
- View product details and seller information.
- Add products to a cart or submit purchase requests.
- Manage listings and orders.
- Receive notifications about marketplace activities.
- Report inappropriate or prohibited listings.
- Allow administrators to manage users and products.

1.5 1.5 References

1. IEEE 830 Software Requirements Specification Standard.
2. Provided reference SRS document.
3. React.js Documentation.
4. Node.js Documentation.
5. Express.js Documentation.

2 Overall Description

2.1 Product Perspective

Campus Stalls is a standalone web application that provides a specialized marketplace for students. It connects student buyers and sellers in a college-focused environment and supports product discovery, listing management, purchase requests, notifications, and administrative moderation.

The system consists of:

- Student User Interface
- Backend Server
- Marketplace and Order Management Module
- MongoDB Database
- Authentication Module
- Notification Module
- Administrator Panel

2.2 Product Features

Major features include:

- User Registration and Login
- Student Profile and Verification
- Product Listing and Management
- Search and Filtering
- Product Details and Availability
- Cart and Order Request
- Seller Dashboard
- Buyer Dashboard
- Notifications
- Reviews and Reporting
- Administrator Dashboard

2.3 User Classes and Characteristics

2.3.1 Student Buyer

- Registers and logs in using valid credentials.
- Searches and filters products.
- Views product and seller information.
- Submits purchase requests.
- Tracks order status.

2.3.2 Student Seller

- Creates and manages product listings.
- Provides product details, price, condition, and images.
- Receives and manages buyer requests.
- Updates listing availability.

2.3.3 Administrator

- Manages student accounts and marketplace listings.
- Reviews reported content.

- Removes prohibited or inappropriate listings.
- Monitors marketplace activity.

2.3.4 System Administrator

- Maintains database and server environment.
- Manages backups and recovery.
- Monitors system security and performance.

2.4 2.4 Operating Environment

Client Side

- Windows, Linux, or macOS
- Google Chrome, Mozilla Firefox, or Microsoft Edge
- Desktop, laptop, tablet, or smartphone

Server Side

- Node.js
- Express.js
- MongoDB
- REST API

2.5 2.5 Design and Implementation Constraints

- Authentication is mandatory for listing and ordering.
- Only permitted college-related products shall be listed.
- Product images shall use JPG, JPEG, or PNG formats.
- MongoDB shall be the primary database.
- HTTPS shall be used for secure communication.
- Users are responsible for accurate listing information.
- The system depends on internet connectivity.

2.6 2.6 User Documentation

The system provides:

- Student User Manual
- Seller/User Guide
- Administrator Manual
- Installation Guide
- API Documentation

2.7 2.7 Assumptions and Dependencies

Assumptions

- Users provide accurate profile and product information.
- Users have stable internet connectivity.
- Students use the platform for legitimate college needs.
- Sellers update product availability when required.

Dependencies

- React.js
- Node.js and Express.js
- MongoDB
- JWT authentication

- Image/file storage
- Web browser and internet

3 3 System Features

3.1 3.1 User Registration and Login

Priority: High

- **REQ-1:** Users shall register using name, student email, phone number, and password.
- **REQ-2:** The system shall validate unique email addresses.
- **REQ-3:** Passwords shall be securely hashed.
- **REQ-4:** Registered users shall log in using valid credentials.
- **REQ-5:** The system shall create an authenticated session/token after successful login.
- **REQ-6:** Users shall be able to log out securely.
- **REQ-7:** Unauthorized access shall be prevented.

3.2 3.2 Student Profile and Verification

Priority: High

- **REQ-8:** Users shall create and update profiles.
- **REQ-9:** The system shall store required profile details.
- **REQ-10:** The system shall support student verification.
- **REQ-11:** Verified status shall be maintained.
- **REQ-12:** Users shall not impersonate other accounts.

3.3 3.3 Product Listing

Priority: High

- **REQ-13:** Sellers shall enter product name, category, description, price, condition, and availability.
- **REQ-14:** Sellers shall upload one or more product images.
- **REQ-15:** Mandatory listing fields shall be validated.
- **REQ-16:** Listing details shall be stored in the database.
- **REQ-17:** Sellers shall edit or delete their listings.
- **REQ-18:** Listings shall support Available, Reserved, Sold, and Removed states.
- **REQ-19:** Prohibited products shall not be listed.

3.4 3.4 Product Search and Discovery

Priority: High

- **REQ-20:** Users shall search products by keyword.
- **REQ-21:** Users shall filter products by category.
- **REQ-22:** Users shall filter products by price range.
- **REQ-23:** Users shall filter products by condition.
- **REQ-24:** Users shall filter products by availability.
- **REQ-25:** Matching products shall be displayed.
- **REQ-26:** Products shall support sorting by price or recent listings.

3.5 3.5 Product Details and Availability

Priority: High

- **REQ-27:** Product name, images, description, price, condition, category, and availability shall be displayed.
- **REQ-28:** Appropriate seller information shall be displayed.

- **REQ-29:** Product status shall be displayed.
- **REQ-30:** Unavailable products shall not be ordered.

3.6 3.6 Cart and Order Request

Priority: Critical

- **REQ-31:** Users shall add available products to cart.
- **REQ-32:** Users shall remove items from the cart.
- **REQ-33:** The system shall calculate the order summary.
- **REQ-34:** Users shall submit purchase requests.
- **REQ-35:** Each order shall have a unique identifier.
- **REQ-36:** Sellers shall receive new order notifications.
- **REQ-37:** Sellers shall accept or reject requests.
- **REQ-38:** Buyers shall view order status.
- **REQ-39:** Statuses shall include Pending, Accepted, Rejected, Completed, and Cancelled.
- **REQ-40:** Items shall be marked Sold only after confirmation.

3.7 3.7 Seller Management

Priority: High

- **REQ-41:** Sellers shall have access to a listing dashboard.
- **REQ-42:** Sellers shall view active, reserved, sold, and removed listings.
- **REQ-43:** Sellers shall update product price, description, availability, and images.
- **REQ-44:** Sellers shall view incoming buyer requests.
- **REQ-45:** Sellers shall update order/request status.
- **REQ-46:** Seller activity shall be recorded.

3.8 3.8 Buyer Dashboard

Priority: High

- **REQ-47:** The dashboard shall display active orders.
- **REQ-48:** Users shall view completed and cancelled orders.
- **REQ-49:** Users may view saved or recently viewed products.
- **REQ-50:** Users shall manage profile information.
- **REQ-51:** Users shall view marketplace notifications.

3.9 3.9 Notifications and Communication

Priority: Medium

- **REQ-52:** Users shall be notified about order creation and status changes.
- **REQ-53:** Notifications shall be displayed in dashboards.
- **REQ-54:** Notification history shall be maintained.
- **REQ-55:** The system may provide buyer-seller communication.
- **REQ-56:** Sellers shall receive relevant listing notifications.

3.10 3.10 Administrator Module

Priority: Critical

- **REQ-57:** Administrators shall manage user accounts.
- **REQ-58:** Administrators shall review and moderate listings.
- **REQ-59:** Administrators shall view reported listings.
- **REQ-60:** Fake, duplicate, or prohibited listings shall be removable.

- **REQ-61:** Administrators shall manage categories.
- **REQ-62:** Administrators shall monitor marketplace activity.
- **REQ-63:** Administrators shall suspend violating accounts.
- **REQ-64:** Administrator actions shall be logged.

3.11 3.11 Reviews and Reporting

Priority: Medium

- **REQ-65:** Eligible buyers shall submit ratings or reviews after completed transactions where enabled.
- **REQ-66:** Users shall report suspicious, misleading, prohibited, or inappropriate listings.
- **REQ-67:** Reports shall be stored for administrator review.
- **REQ-68:** Administrators shall resolve or dismiss reports.
- **REQ-69:** Unauthorized users shall not modify another user's review or report.

4 4 External Interface Requirements

4.1 4.1 User Interface

The system shall provide a simple, responsive, and student-friendly web interface for buyers, sellers, and administrators.

The interface shall provide:

- Registration and Login
- Student Profile
- Marketplace Home Page
- Search and Filter Controls
- Product Details Page
- Create/Edit Listing
- Cart and Order Page
- Buyer Dashboard
- Seller Dashboard
- Notifications
- Reports and Reviews
- Administrator Dashboard

The application shall support desktops, laptops, tablets, and mobile devices.

4.2 4.2 Hardware Interface

Client Requirements

- Desktop, laptop, or smartphone
- Minimum 4 GB RAM recommended
- Stable internet connection
- Camera for product images where supported

Server Requirements

- Intel i5 processor or equivalent recommended
- 8 GB RAM recommended
- MongoDB database server
- Adequate storage for application data

4.3 4.3 Software Interface

The system shall use the following software components:

- React.js
- Node.js
- Express.js
- MongoDB
- JWT authentication
- REST API
- Image upload/storage component

4.4 4.4 Communication Interface

Communication between the client and server shall use REST APIs.

- Protocol: HTTPS
- Data Format: JSON
- Authentication: JWT or secure session
- File Upload: Multipart/form-data

5 5 Other Nonfunctional Requirements

5.1 5.1 Performance Requirements

The system shall:

- Complete normal login requests within approximately 3 seconds under normal network conditions.
- Load product listings within approximately 3 seconds.
- Return search results within approximately 3 seconds for normal dataset sizes.
- Optimize product images for faster loading.
- Support multiple concurrent users.
- Create order records reliably without duplicates.

5.2 5.2 Security Requirements

The system shall provide:

- Secure user authentication.
- Secure password hashing.
- HTTPS communication.
- Role-based access control.
- User-specific listing and profile access.
- Secure file-upload validation.
- Input validation.
- Protection of sensitive information.
- Authorization for administrator functions.

5.3 5.3 Software Quality Attributes

The system shall provide:

- Reliability
- Availability
- Usability
- Maintainability
- Scalability
- Portability
- Efficiency

- Security
- Testability

6 6 Other Requirements

6.1 6.1 Database Requirements

The MongoDB database shall store and manage:

- User and student profile information
- Authentication data
- Product listings
- Product image references
- Categories
- Cart information
- Order/request records
- Notifications
- Reviews and ratings
- Reports and moderation records
- Administrator activity logs

6.2 6.2 Backup and Recovery

The system shall support:

- Regular database backup.
- Recovery after accidental data loss.
- Restoration after system or server failure.
- Backup of critical order and user data.

6.3 6.3 Logging Requirements

The system shall maintain logs for:

- User Login and Logout
- Registration and Profile Updates
- Product Listing Creation
- Product Listing Update or Removal
- Order Creation and Status Changes
- Reports and Moderation Actions
- Administrator Actions
- Authentication or security failures

6.4 6.4 Legal and Ethical Requirements

The system shall:

- Protect user privacy and personal information.
- Inform users about collection and use of information.
- Require users to accept applicable Terms and Conditions.
- Prevent prohibited, illegal, unsafe, or unrelated items.
- Prevent intentional misrepresentation of products.
- Provide mechanisms for reporting misuse.
- Encourage safe and responsible exchanges.

6.5 6.5 Future Enhancements

Possible future improvements include:

- Dedicated Android/iOS mobile application
- Integrated online payment gateway
- Digital student-ID verification
- Campus-location based product discovery
- Wishlist and saved searches
- AI-assisted product recommendations
- Chat between buyer and seller
- QR-code based handover verification
- Email and SMS notifications
- Multilingual support
- Seller analytics and marketplace statistics
- Personalized product recommendations

6.6 6.6 Acceptance Criteria

The Campus Stalls project will be considered successful if:

- Students can register and log in successfully.
- Students can create and manage profiles.
- Verified users can create product listings with required information and images.
- Users can search, filter, and view available products.
- Users can submit purchase/order requests.
- Sellers can view and manage incoming requests.
- Buyers can track order/request status.
- Users receive notifications for important marketplace events.
- Users can report suspicious or inappropriate listings.
- Administrators can manage users, listings, reports, and marketplace activity.
- The system securely stores marketplace data.
- Role-based access is enforced.
- The system operates reliably on supported browsers.

Expected Outcome

Campus Stalls shall provide students with a convenient, organized, and secure platform for discovering, buying, selling, and exchanging college-related products within their student community.

Document Summary

Project Title:

Campus Stalls: A Marketplace Specifically for Students' College Needs

System Type:

Web-Based Student Marketplace

Primary Users:

Students, Buyers, Sellers, Administrators, and System Administrators

Core Modules:

- User Registration and Authentication
- Student Profile and Verification

- Product Listing
- Product Search and Filtering
- Product Details and Availability
- Cart and Order Request
- Seller Management
- Buyer Dashboard
- Notifications and Communication
- Reviews and Reporting
- Administrator Module

Proposed Technology Stack:

- Frontend: React.js
- Backend: Node.js and Express.js
- Database: MongoDB
- Authentication: JWT
- Communication: REST API over HTTPS

Functional Requirements:

The system contains functional requirements REQ-1 through REQ-69, covering authentication, student verification, product listing, search, ordering, seller management, dashboards, notifications, administration, reviews, and reporting.

Nonfunctional Requirements:

- Performance
- Security
- Reliability
- Usability
- Maintainability
- Scalability
- Availability

Expected Outcome:

The completed Campus Stalls system shall provide students with a convenient, organized, and secure marketplace for college needs while supporting controlled student-to-student exchange.